

Kashish Lalwani

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PROFESSIONAL SUMMARY

AI Engineer & Applied Data Scientist bridging technical depth and business impact shipping production ML systems and translating model outputs into measurable decisions. Builds evaluation-first LLM infrastructure, RAG pipelines, and agentic workflows with full ownership from data ingestion and prototyping through optimization and deployment. Drives stakeholder outcomes through KPI design, financial forecasting, and statistical modeling across government, enterprise, and startup environments.

EDUCATION

Stony Brook University – Master of Science in Data Science

Aug 2024 – May 2026

Shivaji University – B.Tech in Computer Science (AI & ML)

Aug 2020 – May 2024

EXPERIENCE

Data Scientist, Stony Brook University, USA (**Retrieval Systems & ML Evaluation**)

Aug 2025 – Dec 2025

- Built SQL/Python pipelines with feature engineering, processing 1M+ records and 100K+ queries for retrieval evaluation.
- Developed True-PIR metrics and bias diagnostics frameworks enabling systematic retrieval robustness assessment at scale.
- Ran root-cause and statistical analysis isolating metric instability, improving decision confidence across stakeholders.
- Set up monitoring pipelines for latency, throughput, and execution metrics, surfacing production performance risk early.
- Implemented validation and audit processes, strengthening data integrity and governance for reliable analytics outcomes.

Founding AI Engineer, Versaunt (fka. Aayats), Inc., USA (**ML Systems, API & Data Pipelines**)

May 2025 – Aug 2025

- Built FastAPI backend services supporting 10K+ users and 500K+ queries, scaling AI inference pipelines across production systems.
- Developed RAG pipelines using LLMs, embeddings, and vector search, increasing retrieval relevance and response accuracy.
- Designed A/B testing frameworks across 50K+ daily events, driving cohort analysis and cross-functional product decisions.
- Engineered real-time data pipelines boosting scalability and consistency, shrinking event processing time in production.
- Reduced inference latency and improved throughput via batching and caching optimizations for high-concurrency production.

Data Analyst, Saroj Group of Companies, IN (**Analytics & Automation**)

Jan 2024 – July 2024

- Applied statistical modeling on 100K+ records, cutting defect rates 40% and improving operational workflow efficiency.
- Led financial forecasting and cost scenario analysis that increased profit margins by 12% through targeted optimization.
- Built KPI dashboards in SQL, Excel, and Tableau, reducing reporting cycle time 35% and improving cross-team analytical accuracy.
- Automated four pipelines in Python and SQL, eliminating 25+ monthly hours of manual effort and accelerating delivery speed.

Data Analyst, Kamala Plastics, IN (**Operations Analytics & Data Quality**)

July 2023 – Dec 2023

- Developed ETL pipelines integrating 200K+ records, enabling robust statistical analysis across all production departments.
- Identified root-cause patterns reducing scrap by 22% and product returns by 15% through targeted process improvements.
- Maintained quality via validation, reconciliation, and standardized metric tracking across all operational data pipelines.
- Used SQL and Python to analyze operational trends, improving production performance by 17% across manufacturing workflows.

PROJECTS

RAG Financial Intelligence Platform (Confidential: Gov. of India)

Sept 2023 – May 2024

Technical Stack: Python, LLMs, LangChain, Vector Search, ETL

- Shipped end-to-end RAG pipelines for a government client, reducing analytical latency 60% and decision efficiency by 40%.

Aegis: Cost-Aware Real-Time Fraud Detection System

Mar 2026 – May 2026

Technical Stack: PySpark, Databricks, Python, MLflow, Drift Detection

- Built PySpark fraud pipeline on Databricks, achieving F1 0.9942 and AUC 0.993 while cutting financial exposure by \$19.4M+.

Structured Clinical Data Pipeline (Harvard HackRare 2026)

March 2026

Technical Stack: Python, spaCy, negspacy, HPO, GA4GH

- Resolved negation and uncertainty in rare disease notes, mapping phenotypes to HPO and GA4GH standards at Harvard HackRare.

AI Portfolio Risk & Forecasting Model

Feb 2025 – May 2025

Technical Stack: Python, XGBoost, CVaR Optimization, Time-Series

- Designed a CVaR-based optimization model on 10+ years of data, raising Sharpe ratio 15% and slashing downside risk by 28%.

SKILLS & INTERESTS

Programming & Databases: Python, SQL, C, PostgreSQL, MySQL, BigQuery, PyTorch, TensorFlow, FastAPI, REST APIs

ML / AI & GenAI: LLMs, RAG, Prompt Engineering, Transformer Models, Agentic Workflow Optimization, Embeddings, Vector Search, FAISS, LangChain, Clinical NLP, Model Optimization, Model Evaluation, Model Deployment, MLOps, Feature Engineering, Language Modeling, Statistical Testing, Hypothesis Testing, Predictive Modeling, Benchmarking, Hyperparameter Tuning, Cross-Validation

Analytics & BI: KPI Design, A/B Testing, Cohort Analysis, Forecasting, Root-Cause Analysis, Data Storytelling, Data Validation, Scenario Analysis, Financial Forecasting, P&L Analysis, Excel (Advanced), Tableau, Power BI, Looker, Snowflake

Systems & Production ML: ETL/ELT, Data Pipelines, CI/CD (Git), Inference Optimization, Production Monitoring, Governance, Latency & Throughput Optimization, Big Data, PySpark, Pandas, Scikit-learn, Databricks, AWS, MLflow

Interests: Financial Investments, Financial Markets, Professional Chess, Music

LEADERSHIP, AWARDS & RECOGNITION

Best Outgoing Student (sole recipient) | Student Council Secretary (3x) | Smart India Hackathon Finalist — Ministry of Power (National)

| Rotaract: Best Emerging Rotaractor across 75+ clubs, District 3170 Secretary of the Year | District Chess Champion (2023 & 2024)